

Motion hides in the timing

Exercise session 2 · Read the picture, find the unknowns

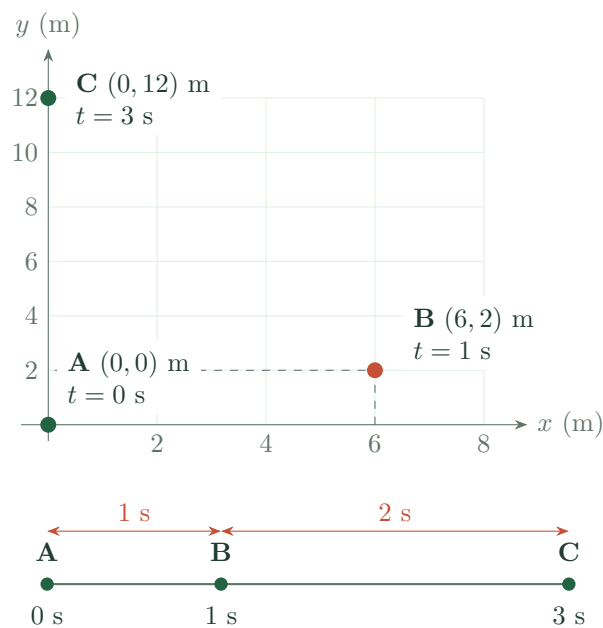
Problems 1 and 2 are required; Problem 3 is an optional challenge.

PROBLEM 1

Three snapshots, one missing story

A robot is moving. You have three camera snapshots. The same constant acceleration throughout.

Reconstruct the motion: $\vec{v}_0 = ?$, $\vec{a} = ?$



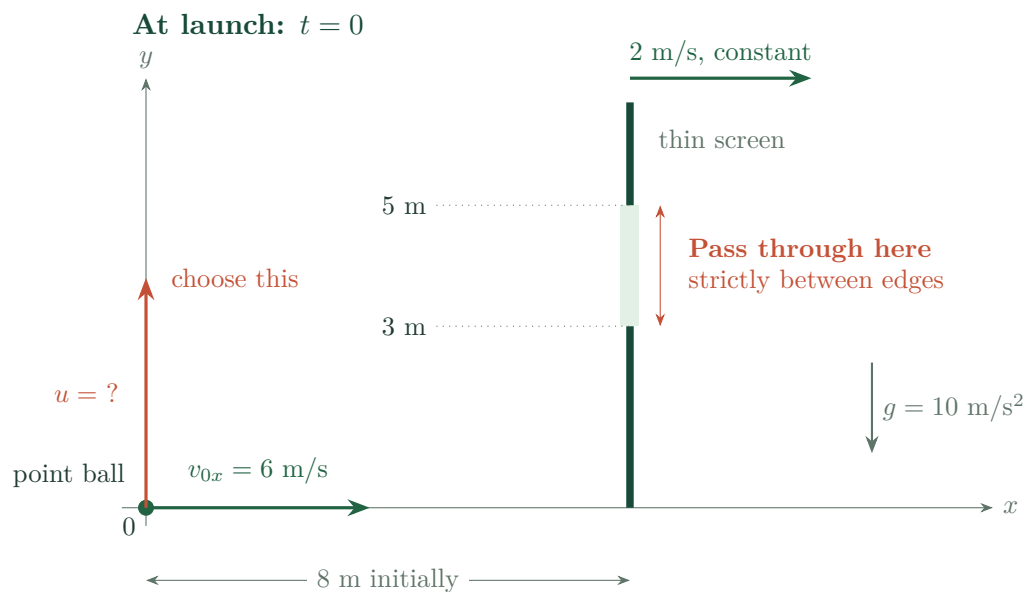
- Find the initial velocity $\vec{v}_0$ and acceleration $\vec{a}$.
- When is the robot farthest right? Is it stopped at that instant?
- Find its average velocity from A to C.

Working and explanation

PROBLEM 2

A throw through a moving window

Choose the upward launch component u so the ball clears the moving window.



No air resistance. The screen keeps moving during the flight.

- When does the ball reach the screen?
- What range of u lets it pass through the window?
- To pass through the centre, what must u be? Is the ball rising or falling there?

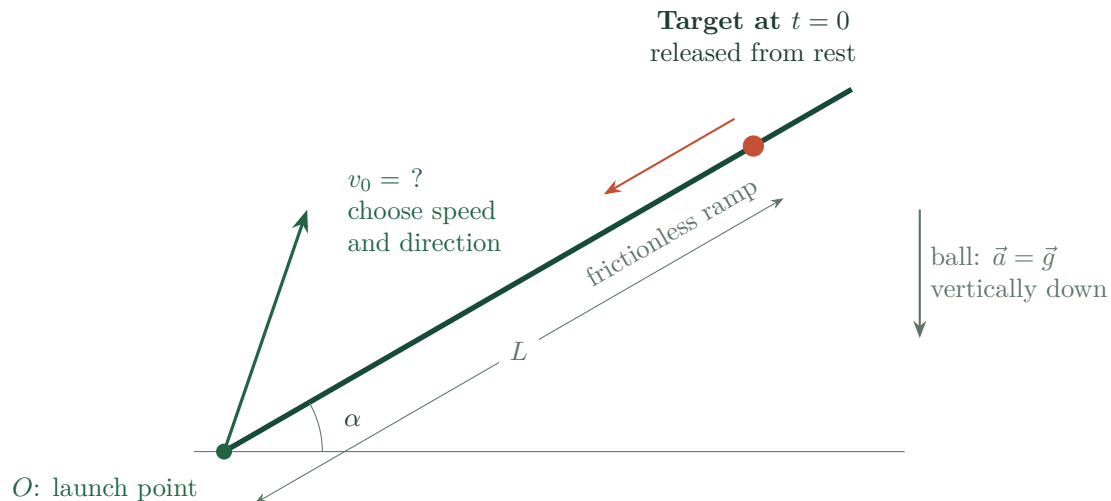
Working and explanation

PROBLEM 3

The slowest possible interception

Optional · Difficult

Release the target and launch the ball at the same instant. Find the slowest successful launch. (How would you choose the coordinate system so that motion is simplest?)

**Successful interception**

Meet on the ramp at $t > 0$, before the target reaches O .

Both objects are points. $0 < \alpha < 90^\circ$. No air resistance.

- For a meeting at time t , find the launch components v_{0s} and v_{0n} , and the allowed range of t .
- Ignoring the deadline at O , find the least launch speed and its direction.
Hint: use $(A - B)^2 \geq 0$.
- For which ramp angles is that minimum possible? For the others, find the speed that can be approached but never attained.

Working and explanation
